Distinct roots and the derivative
Axler, 4.4

Theorem 1. *Suppose p is a nonzero polynomial of degree m with complex coefficients. If p and its derivative p' have no root in common, then p has m distinct roots. In other words, there exists a finite set $S \subset \mathbb{C}$ such that $|S| = m$ and every $z \in S$ is a root of p.*

Proof. Since $\mathbb{C}$ is algebraically closed, p splits over $\mathbb{C}$. Let $\operatorname{roots}(p)$ denote the multiset of roots of p, in which each root occurs according to its multiplicity.

We first show that $\operatorname{roots}(p)$ has no repeated element. Suppose, toward a contradiction, that some $a \in \mathbb{C}$ occurs more than once in $\operatorname{roots}(p)$. The number of occurrences of a in $\operatorname{roots}(p)$ is the multiplicity of a as a root of p, so this multiplicity is at least 2.

Since $(X - a)^k$ divides p, where k is the multiplicity of a as a root of p, and since that multiplicity is at least 2, it follows that

$$(X - a)^2 \mid p.$$

Thus there is a polynomial q with complex coefficients such that

$$p = (X - a)^2 q.$$

In particular, a occurs in $\operatorname{roots}(p)$, so a is a root of p.

We show that a is also a root of p'. By the product rule and the rule for differentiating a power,

$$p' = 2(X - a)q + (X - a)^2 q'.$$

Evaluating at a gives

$$p'(a) = 2(a - a)q(a) + (a - a)^2 q'(a) = 0.$$

Hence a is a common root of p and p', contradicting the hypothesis that p and p' have no root in common. Therefore $\operatorname{roots}(p)$ has no repeated element.

Because p splits over $\mathbb{C}$, the number of roots of p, counted with multiplicity, equals the degree of p. Thus

$$|\operatorname{roots}(p)| = \deg p = m.$$

Let S be the underlying set of the multiset $\operatorname{roots}(p)$. Since this multiset has no repeated element, S has the same number of elements as the multiset, so

$$|S| = |\operatorname{roots}(p)| = m.$$

Finally, every element of S lies in $\operatorname{roots}(p)$ and hence is a root of p. Therefore S is a set of m distinct roots of p, as required. □